SUPPORT AND SHEAFINESS PROPERTIES OF THE RIGHT ADJOINT FOR A PROPER FLAT MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0E4J (sources/stacks-duality/pinned-original-source.tex line 2866).

Allowed standard prerequisites (editorial).

- The sheafyness criterion for the right adjoint. Same chapter, Lemma labelled lemma-when-sheafy, cited at proof lines 2882-2884. Gives the isomorphism of part (2).
- Support of the right adjoint in the Noetherian case. Same chapter, Lemma labelled lemma-proper-noetherian, cited at proof lines 2882-2884. Gives part (1) when the complement of T is quasi-compact open.
- Commutation with direct sums for proper flat morphisms. Same chapter, Lemma labelled lemma-proper-flat, cited at proof lines 2882-2884. Supplies the flatness input used by the other two lemmas.

Source reference. Original lemma 0A9E: sources/stacks-duality/pinned-original-source.tex, source lines 653–653. Same-chapter pointer into the author environment lemma at source line 653; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original equation 0A9L: sources/stacks-duality/pinned-original-source.tex, source lines 988–988. Same-chapter pointer into the author environment equation at source line 988; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A9N: sources/stacks-duality/pinned-original-source.tex, source lines 1035–1035. Same-chapter pointer into the author environment lemma at source line 1035; invoked at proof line 2883 of the counted proof.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A9P: sources/stacks-duality/pinned-original-source.tex, source lines 1106–1106. Same-chapter pointer into the author environment lemma at source line 1106; invoked at proof line 2884 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0E4I: sources/stacks-duality/pinned-original-source.tex, source lines 2831–2831. Same-chapter pointer into the author environment lemma at source line 2831; invoked at proof line 2885 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let Y be a quasi-compact and quasi-separated scheme. Let $f : X \to Y$ be a morphism of schemes which is proper, flat, and of finite presentation. Let a be the right adjoint for $Rf_* : D_{QCoh}(\mathcal{O}_X) \to D_{QCoh}(\mathcal{O}_Y)$ of Lemma 0A9E. Then*

1. *for every closed $T \subset Y$ if $Q \in D_{QCoh}(Y)$ is supported on T, then $a(Q)$ is supported on $f^{-1}(T)$,*
2. *for every quasi-compact open $V \subset Y$ and any $K \in D_{QCoh}(\mathcal{O}_Y)$ the map (0A9L) is an isomorphism.*

Proof. This follows from Lemmas 0A9N, 0A9P, and 0E4I. Strictly speaking part (1) follows only for those T whose complement is a quasi-compact open; however the general case then follows as any closed subset is an intersection of such. □